Voltage Protector Device Design Report

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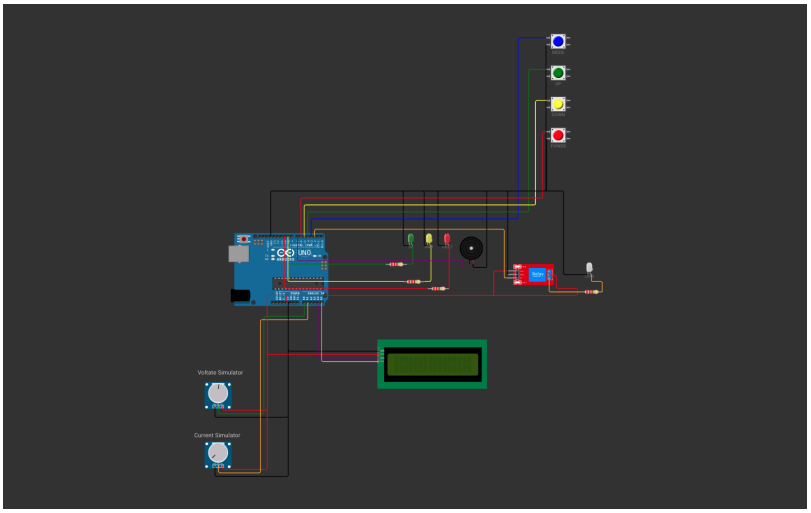


Figure 1: Wokwi circuit layout showcasing the Arduino Uno connected to dual potentiometer sensor simulators, interactive LCD display, configuration pushbuttons, status LEDs, and relay load control.

1. Conceptualization & Problem Statement

When I first reviewed the question, I immediately connected it to household surge protectors and realized the core problem. I knew the system required three fundamental components—ADCs to sample current and voltage levels, and a programmable switch or relay to disconnect the load during faults. Given the tight time constraints and impossibility of procuring physical sensors within a single day, I decided to build and test the design using an online simulator, choosing Wokwi for its accessibility and visual debugging capabilities.

2. Circuit Design & Implementation Strategy

Because Wokwi lacks specialized dedicated voltage and current sensor modules, I pivoted to using two potentiometers to simulate real-time analog input ranges for voltage and current. After validating basic relay actuation logic, I faced another constraint: difficulty establishing a reliable simulated computer serial connection. To overcome this, I integrated four physical pushbuttons directly into the circuit alongside an LCD screen, allowing local threshold configuration and navigation. Once the wiring structure was finalized, I combined my engineering foundation with AI assistance to develop, test, and polish the firmware into a functional prototype.